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## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING ENGINEERING

## " APPLIED ARTIFICIAL INTELLIGENCE: FROM DATA ETHICS AND BIAS MITIGATION TO DEEP LEARNING AND VOICE COMMERCE SYSTEMS "

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# MODULE 2
# DATA ETHICS CASE STUDY

***Question (Module 2):*** *Data Ethics Case Study: Each group analyses a real or hypothetical case of data misuse (like the Cambridge Analytica case) and discusses how better governance and ethical practices could have prevented it.*

## *2.1 Introduction and Case Selection*

Data ethics is the branch of applied ethics that examines how information about people is collected, shared, analysed, and used, and that asks whether those practices respect autonomy, privacy, fairness, and accountability. As organisations accumulate ever larger datasets, the gap between what is technically possible and what is ethically acceptable has widened, and a single lapse can now affect millions of people at once. The group therefore chose a real and widely documented case of data misuse, the Facebook and Cambridge Analytica affair of 2014 to 2018, and analysed it in three steps: what actually happened, which ethical and governance principles were broken, and how better governance could have prevented the harm.

The case was selected for three reasons. It is well documented in investigative journalism, regulatory findings, and legal settlements, so the facts can be checked rather than assumed. It involved ordinary consumers who never knowingly agreed to the use that was eventually made of their data. And it produced concrete regulatory consequences, which makes it possible to compare what happened with what a sound governance framework would have required. Where this report simplifies a complicated story, the simplification is stated openly so that the analysis remains honest and the figures quoted can be traced to public reporting.

## *2.2 Background and Timeline of the Case*

Until 2014 the Facebook platform allowed outside developers to build applications that users could log into with their Facebook accounts. When a user approved such an application, it could often read not only the user's own profile information but also information about the user's friends, who had never been asked. In 2013 and 2014 an academic researcher, Aleksandr Kogan, built a personality quiz of this kind through his company Global Science Research. The quiz was presented as research, participants were paid a small sum, and the information gathered was later passed to the political consulting firm Cambridge Analytica, which had commercial and political clients in several countries. Table 2.1 summarises the sequence of events as it was later reported.

***Table 2.1:*** *Timeline of the Facebook and Cambridge Analytica case (as publicly reported)*

| Period | Event |
|---|---|
| 2013–2014 | A quiz application built by Global Science Research collects data from participants and, through platform permissions, from their friends. |
| 2014 | Around 270,000 people install the application; the data reaches a far larger network of friends who never used it. |
| 2015–2016 | Press reports reveal the transfer to Cambridge Analytica; Facebook asks for the data to be deleted and accepts written assurances. |
| March 2018 | The Guardian, The Observer, The New York Times, and Channel 4 News publish detailed reports based on a former employee's account. |
| April–May 2018 | Facebook states that up to 87 million users may have been affected; its chief executive testifies before the US Congress; Cambridge Analytica begins insolvency proceedings. |
| 2018–2019 | The UK regulator fines Facebook 500,000 pounds, and the US Federal Trade Commission announces a 5 billion US dollar penalty. |

## *2.3 How the Data Was Collected and Misused*

The harvesting depended on the permissions model of the platform. Participants who logged into the quiz agreed to share their profile, their “likes”, and in many cases their friend lists, and the application could then retrieve a comparable set of information about those friends. Combining the quiz scores with the pages that people had liked allowed researchers to train models that estimate personality traits from online behaviour. Those models could then be applied to anyone whose likes were known, which is why data from a few hundred thousand participants could be extended to a much larger population. Platform rules permitted collection for academic purposes but forbade selling or transferring the data to others, and it was this rule that was broken.

Cambridge Analytica used the resulting profiles to build psychographic segments, matched them with voter records, and designed messages intended to appeal to the traits of each segment. The extent to which such micro-targeting changed voting behaviour is disputed, and this report does not claim that it did. The ethical problem lies in the way the data was obtained and used: people were profiled for a political purpose they had never been told about, and most of them had not given consent of any kind. Figure 2.1 traces the path of the data from collection to use and shows how disclosure in 2018 created the pressure for reform.

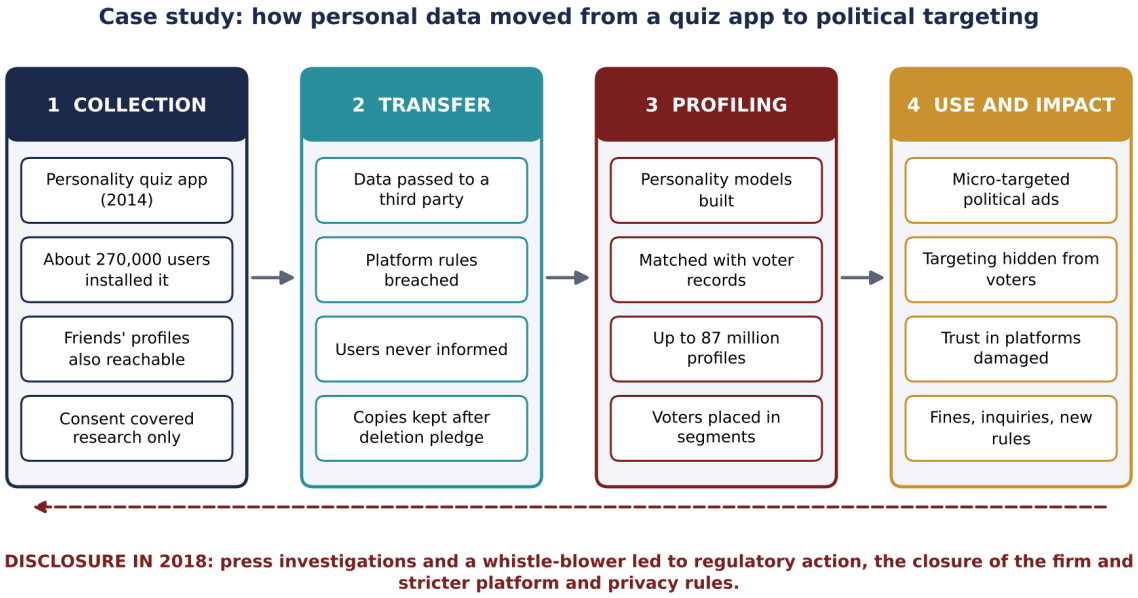



***Figure 2.1:*** *Flow of personal data in the case study, from the quiz application to political targeting, with the consequences of disclosure shown as a feedback loop.*

## *2.4 Ethical Principles and Governance Failures*

Seen through the main principles of data protection, the case contains several separate failures rather than a single mistake. Consent was missing for the friends whose data was collected and was too narrow for the participants, who had agreed to take part in a research quiz, not to supply a political consultancy. Purpose limitation was ignored because data gathered for one stated purpose was reused for another. Transparency was lacking because neither participants nor friends were told how their data would be used. Data minimisation was not applied, since the application collected far more than a personality quiz required. Finally, accountability failed on the platform side: when the transfer was discovered, Facebook relied on written assurances that the data had been deleted instead of verifying it by audit.

***Table 2.2:*** *Ethical and governance principles mapped to the failures observed in the case*

| Principle | What the case shows | Effect on people |
|---|---|---|
| Informed consent | Friends were never asked; participants agreed to research use only. | Loss of control over personal data. |
| Purpose limitation | Academic data was reused for political profiling. | Data used against the expectations of its owners. |
| Transparency | No clear notice of onward transfer or profiling. | Users could not object or withdraw. |
| Data minimisation | Friend networks and likes collected well beyond the quiz's needs. | Larger exposure when the data was misused. |
| Accountability | Deletion was certified on paper but not independently checked. | Misuse continued undetected for years. |

Beyond the individual principles, the case raised a wider concern about autonomy and democratic fairness. Targeted messages that are visible only to the people receiving them cannot be examined or challenged by others, which weakens public debate. This is why data ethics treats the collective effects of data use as seriously as the effects on any single person.

## *2.5 How Better Governance Could Have Prevented the Misuse*

Governance can act at three levels, and a failure at any one of them was enough to let the misuse happen. At the platform level, privacy by design would have meant that applications could not read friends' data by default, that every permission requested had to be justified, and that apps were reviewed before and after launch. Technical controls such as limits on the volume of data an application may retrieve, together with monitoring for unusual bulk collection, would have made harvesting on this scale far harder. Where a violation was found, verified deletion backed by an independent audit would have replaced reliance on a signed statement.

At the organisational level, companies that handle personal data need clear stewardship. A data protection impact assessment before any new use, an ethics review for projects that profile people, contracts that give the data owner the right to audit partners, and a protected channel through which employees and researchers can raise concerns are all inexpensive compared with the harm they prevent. At the regulatory level, laws such as the European General Data Protection Regulation and, in India, the Digital Personal Data Protection Act of 2023 require lawful and specific consent, notification of breaches, and penalties large enough to change behaviour. For users, easy-to-find privacy dashboards and one-step withdrawal of consent turn rights on paper into rights in practice. Figure 2.2 places these controls at the stage of the data lifecycle where they would have intervened.

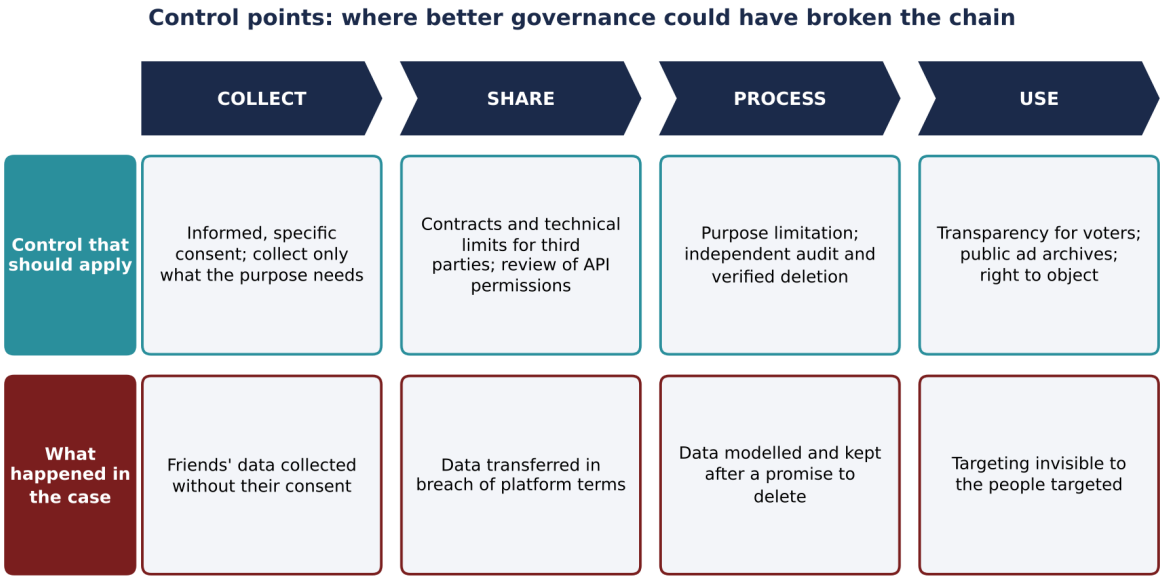



***Figure 2.2:*** *Control points across the data lifecycle, set against what actually happened in the case.*

## *2.6 Consequences, Lessons and Conclusion*

The consequences of the affair were wide. Public trust in large platforms fell sharply, regulators in several countries opened inquiries, Cambridge Analytica closed, and Facebook paid financial penalties, restricted the data available to outside applications, and agreed to a stronger privacy programme under the supervision of the US regulator. The case also helped shape the public debate that led to stricter privacy laws and to wider acceptance that data protection is a matter of public interest as well as private concern.

The main lesson of the group's analysis is that ethical data practice cannot rest on good intentions or on policies that exist only on paper. Consent must be informed and specific, data must be used only for the purpose stated, partners must be audited, and responsibility must be assigned to named people and enforced by law. Better governance would not have required new technology; it would have required that rules already written be applied and verified. To test whether these lessons apply beyond one famous example, the group also considered a hypothetical case: a fitness-tracking application that shares its users' location and heart-rate data with an insurance partner so that premiums can be adjusted. The same questions arise. Did users consent to this specific use, is the data limited to what is needed, can they withdraw, and who checks the partner? The answers expose the same gaps in consent, purpose limitation, and accountability, and the same controls would address them. In short, the group recommends a simple checklist for any data project: state the purpose, obtain specific consent, collect the minimum, restrict and audit sharing, set a deletion date, and name the person who is accountable. The next module extends this reasoning from how data is collected to how it is used by machine learning models, where similar failures appear as bias.

# MODULE 3
# ML ETHICS AND BIAS CASE STUDY

***Question (Module 3):*** *ML Ethics & Bias Case Study: Groups analyze a case where machine learning caused bias or unintended consequences (like biased hiring tools or unfair credit scoring) and propose solutions or guidelines to improve the model.*

## *3.1 Introduction and Case Selection*

Machine learning systems learn patterns from historical data and then apply them to new people, which means that any unfairness present in the past can be learned, repeated, and even amplified at scale. Bias in this sense is a systematic error that disadvantages some groups of people, and it can arise even when no one intends it and when the protected attribute, such as gender, has been removed from the data. The group chose the experimental recruiting tool reported by Reuters in 2018 as its case study, because it shows clearly how a model that is accurate with respect to past decisions can still be unfair to the people it judges.

Hiring and credit scoring are the two classic domains for this problem, since both decide who receives an opportunity and both rely on historical records. The analysis follows the same structure as the previous module: what happened, why it happened, how it can be measured, and what guidelines would make the model safer.

## *3.2 The Case: A Recruiting Model That Learned to Prefer Men*

According to the reporting, the company began building an automated tool in 2014 that rated job applicants on a scale of one to five stars so that recruiters could identify promising candidates quickly. The model was trained on résumés submitted to the company over about ten years. Because the technology industry has been dominated by men, most of those résumés came from men, and past hiring decisions reflected the same imbalance. The model learned that features common among men were associated with being hired. It reportedly downgraded résumés containing the word “women’s”, as in a women’s chess club, and penalised graduates of two all-women’s colleges.

Engineers edited the system to treat those particular terms neutrally, but they could not be sure that the model would not find other proxies for gender, and the project was reportedly abandoned. The case is valuable precisely because nobody told the model to discriminate. It did exactly what it was built to do, which was to imitate past decisions, and the past decisions were biased. Credit scoring models show the same mechanism when they rely on features such as postcode, which can stand in for caste, religion, or race and so reproduce historic patterns of exclusion.

The harm in such cases is practical as well as moral. A qualified candidate who is screened out never learns why, a lender who relies on the score may unknowingly break anti-discrimination law, and the organisation loses talent and public trust. Because these systems operate at scale, a small bias in the model becomes a large pattern of unequal outcomes, which is why the group treated the recruiting tool as a warning about every automated decision system and not as an isolated failure.

## *3.3 Sources of Bias in the Machine Learning Pipeline*

Bias can enter at every stage of the pipeline, and it is therefore not enough to inspect the model alone. Historical bias is present when the world itself was unequal at the time the data was collected. Representation bias arises when some groups are under-represented in the training data. Measurement bias appears when the label or the feature is a poor measure of what really matters, such as treating "hired in the past" as a proxy for "good employee". Aggregation and evaluation bias occur when a single model is tested only on the majority group, and feedback bias occurs when the model's own decisions shape the data on which it is retrained. Figure 3.1 shows where these biases enter the hiring model and where each can be reduced.

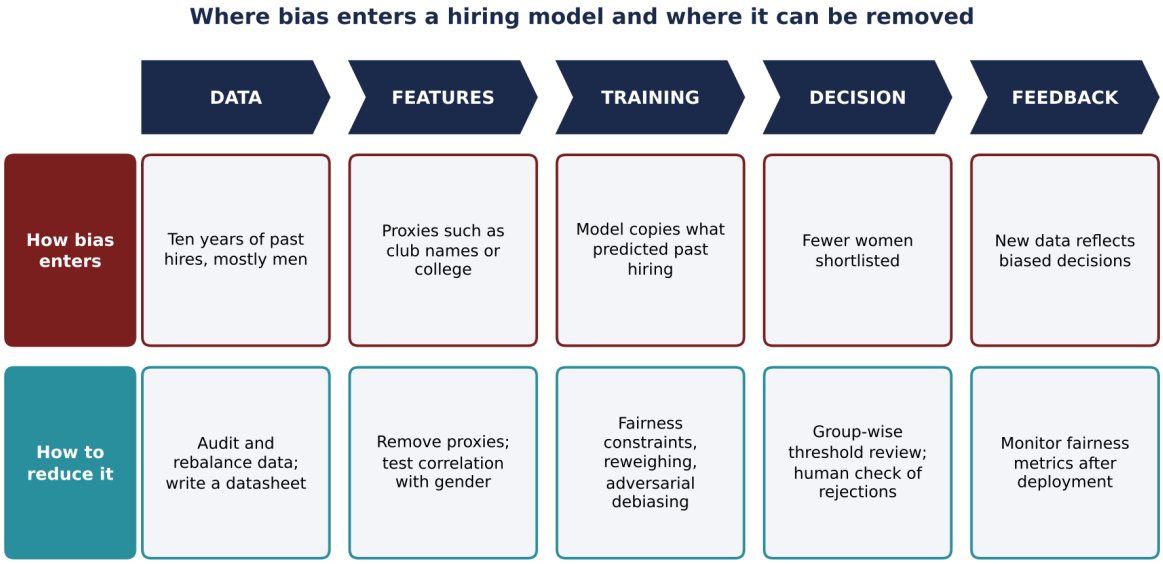



***Figure 3.1:*** *Stages of a hiring model at which bias enters, with a mitigation for each stage.*

***Table 3.1:*** *Types of bias and how each appeared in the recruiting case*

| Type of bias | Meaning | Appearance in the case |
|---|---|---|
| Historical bias | The world was unequal when data was gathered. | Ten years of mostly male hires. |
| Representation bias | Some groups are scarce in training data. | Few résumés from women in technical roles. |
| Proxy bias | A neutral feature stands in for a protected one. | Club names, college names, and word choice. |
| Measurement bias | The label is a poor measure of the goal. | Past hiring used as proof of job quality. |
| Feedback bias | Model outputs shape future training data. | Fewer women shortlisted, so fewer hired, so less data. |

## *3.4 Measuring Fairness*

A bias that cannot be measured cannot be managed, so fairness must be translated into metrics. Statistical parity compares the rate at which each group receives the favourable outcome, and the disparate impact ratio divides the lower rate by the higher one. The four-fifths rule used in United States employment guidance treats a ratio below 0.80 as a warning sign. Equal opportunity compares true positive rates, asking whether qualified candidates from every group are equally likely to be selected, while predictive parity asks whether a positive prediction means the same thing for every group. Research has shown that these definitions cannot all be satisfied at once when base rates differ between groups, so the team has to choose the metrics that fit the harm that matters most and document the reason. For a hiring tool the group considered equal opportunity the most important, since the central harm is that a qualified woman is less likely to be shortlisted than an equally qualified man, and it recommended that statistical parity be reported alongside it as a safeguard.

Table 3.2 and Figure 3.2 illustrate the idea with hypothetical numbers for 1,000 applicants, 600 men and 400 women. The values are invented for demonstration and are not data from any company. Before mitigation the model selects 40 per cent of men but only 20 per cent of women, a ratio of 0.50. After the mitigation described in the next section the ratio rises to 0.89, at a small cost in predictive quality.

***Table 3.2:*** *Hypothetical fairness audit of a screening model before and after mitigation (illustrative values)*

| Measure | Before | After |
|---|---|---|
| Men selected (of 600) | 240 (40%) | 216 (36%) |
| Women selected (of 400) | 80 (20%) | 128 (32%) |
| Statistical parity difference (women minus men) | −20 points | −4 points |
| Disparate impact ratio | 0.50 | 0.89 |
| Predictive quality (illustrative AUC) | 0.84 | 0.82 |

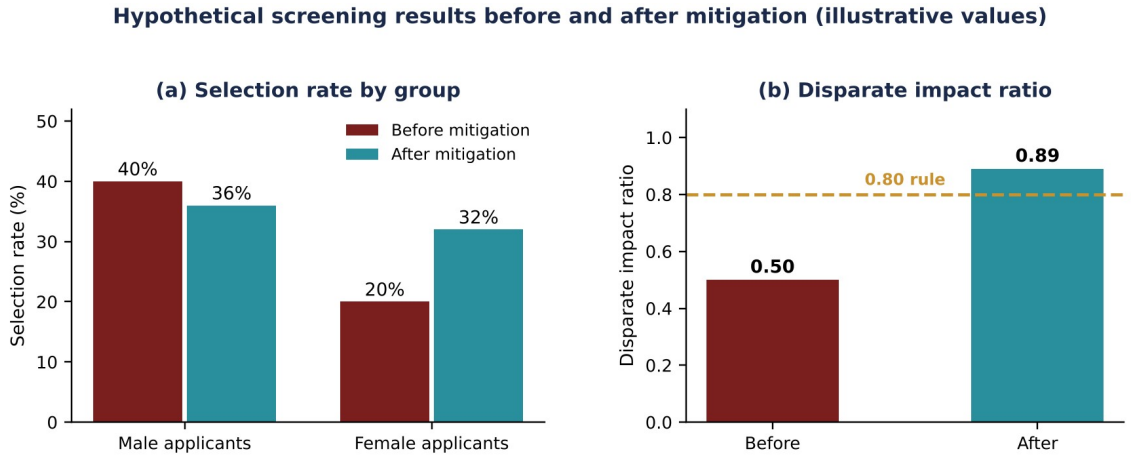



***Figure 3.2:*** *(a) Selection rate by group and (b) disparate impact ratio before and after mitigation, with the four-fifths threshold of 0.80.*

## *3.5 Proposed Solutions and Guidelines*

Technical solutions fall into three families. Pre-processing methods repair the data before training, for example by reweighing examples so that groups are balanced, collecting more representative data, and removing proxy features that correlate strongly with the protected attribute. In-processing methods change the learning objective by adding a fairness constraint or an adversarial component that penalises the model if the protected attribute can be predicted from its output. Post-processing methods adjust the decision threshold for each group after training so that error rates are balanced. The hypothetical improvement in Table 3.2 corresponds to reweighing combined with a threshold review.

Technology alone is not sufficient, so the group proposes a set of guidelines for the redesigned system. First, define fairness goals and the affected groups before development begins, and record them. Second, document datasets and models with datasheets and model cards, which state the intended use, the training data, and known limitations. Third, test the model on each group separately and publish the results of the audit. Fourth, keep a human reviewer in the loop for rejections, with a clear route for candidates to appeal. Fifth, monitor the deployed model for drift in fairness metrics and retrain or withdraw it when thresholds are crossed. Sixth, involve legal, domain, and community stakeholders in review, and comply with applicable anti-discrimination and data protection law.

***Table 3.3:*** *Guidelines proposed for the redesigned recruiting model*

| Stage | Guideline | Evidence of compliance |
|---|---|---|
| Planning | Define fairness goals and the affected groups before development. | Written fairness requirement. |
| Data | Audit representation and remove proxy features. | Datasheet for the training set. |
| Training | Apply reweighing or a fairness constraint and compare with a baseline. | Model card with group-wise results. |
| Testing | Report selection rates and error rates for every group. | Audit report reviewed by an independent team. |
| Deployment | Keep human review of rejections and a route for appeal. | Documented review and appeal process. |
| Monitoring | Track fairness metrics on live data and retrain or withdraw if thresholds are crossed. | Monthly fairness dashboard. |

## *3.6 Conclusion*

The recruiting case shows that a model can be technically successful and ethically wrong at the same time, because it learns the prejudices of the data it is given. Bias must therefore be looked for at every stage, measured with explicit metrics, and reduced through a mixture of data repair, modified training, and sound organisational process. The guidelines proposed here do not make a model perfectly fair, since the definitions of fairness themselves conflict, but they make unfairness visible, accountable, and correctable, which is the realistic goal of ethical machine learning.

# MODULE 4
# ARCHITECTURE COMPARISON DEBATE

***Question (Module 4):*** *Architecture Comparison Debate: Split into groups; each defends RNN, CNN, or GAN—covering structure, use-cases, strengths, and weaknesses.*

## *4.1 Introduction and Debate Format*

Deep learning architectures differ in how they connect their neurons, and that connection pattern decides which kinds of data they handle well. For the debate the class was divided into three groups, and each group defended one architecture: the Recurrent Neural Network (RNN), the Convolutional Neural Network (CNN), or the Generative Adversarial Network (GAN). Every group was asked to cover the same four points, namely the structure of the network, the use cases in which it is strongest, its main strengths, and its weaknesses, so that the arguments could be compared fairly at the end. The three structures are shown side by side in Figure 4.1.

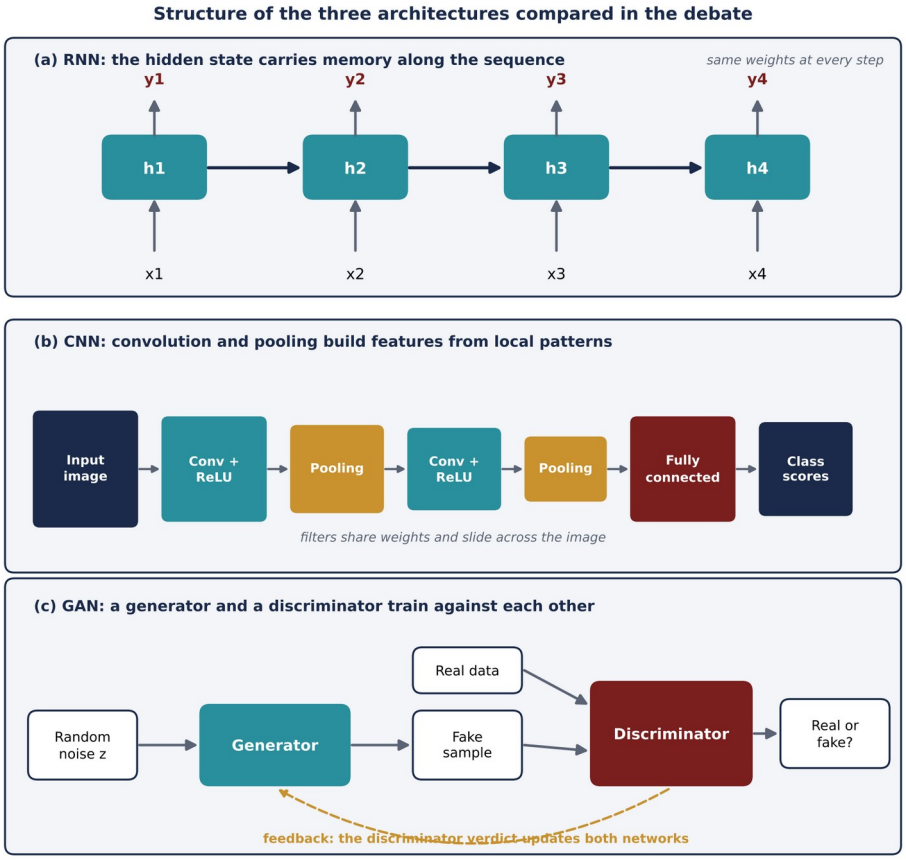



***Figure 4.1:*** *Structure of (a) a recurrent neural network unrolled over four time steps, (b) a convolutional neural network, and (c) a generative adversarial network.*

## *4.2 Recurrent Neural Networks*

A recurrent neural network processes data one element at a time and carries a hidden state from each step to the next, so that its output at any moment depends on the current input and on a summary of everything seen before. The same weights are reused at every step, which allows a single network to handle sequences of any length. The plain form of the network is difficult to train over long sequences because the gradient shrinks or grows as it is propagated back through many steps, and this problem led to the Long Short-Term Memory (LSTM) unit of Hochreiter and Schmidhuber in 1997 and to the simpler Gated Recurrent Unit (GRU) introduced in 2014, both of which use gates to control what is remembered and what is forgotten.

RNNs are the natural choice for data in which order matters: speech recognition, handwriting recognition, language modelling, machine translation, music generation, and forecasting of time series such as demand, weather, or sensor readings. Their strengths are an explicit memory, a compact model that does not grow with sequence length, and good performance on moderate-length sequences with limited data. Their weaknesses are that computation must proceed step by step, which prevents parallel training and makes them slow on modern hardware, that even LSTMs struggle with very long dependencies, and that attention-based Transformer models have now replaced them in most large language tasks.

## *4.3 Convolutional Neural Networks*

A convolutional neural network is designed for data arranged on a grid, most commonly an image. Instead of connecting every input pixel to every neuron, it slides small learnable filters across the image, so that each filter detects a local pattern such as an edge or a texture wherever it appears. Pooling layers then reduce the size of the feature maps and make the representation tolerant of small shifts, and later layers combine simple patterns into parts and finally whole objects before a fully connected layer produces class scores. The idea was demonstrated by LeNet-5 for digit recognition in 1998 and became dominant after AlexNet won the ImageNet competition in 2012 by a wide margin.

CNNs are used for image classification, object detection, face recognition, medical image analysis, document and number-plate reading, and, in one-dimensional form, for audio and text. Their strengths come from weight sharing and local connectivity, which sharply reduce the number of parameters, from automatic feature learning that removes the need for hand-designed features, and from highly parallel computation on graphics processors. Their weaknesses are a need for large labelled datasets, limited ability to capture relationships between distant parts of an image, sensitivity to rotation and scale unless augmented, vulnerability to adversarial perturbations, and limited interpretability.

Later designs show how far the idea has been extended. VGG networks demonstrated that stacking many small filters improves accuracy, and residual networks (ResNet), introduced in 2015, added shortcut connections that allow networks with more than a hundred layers to be trained without the gradient vanishing. Today CNN backbones are also embedded inside object detectors and segmentation models, and the group argued that this long record of reuse is the strongest single point in favour of the architecture: a trained CNN is easy to adapt through transfer learning to a new task with only a small amount of new data.

## *4.4 Generative Adversarial Networks*

A generative adversarial network, proposed by Goodfellow and colleagues in 2014, consists of two networks trained in competition. The generator takes random noise and produces synthetic samples, while the discriminator receives a mixture of real and generated samples and tries to tell them apart. The two are trained in turn: the discriminator improves at detecting fakes, and the generator uses the discriminator's feedback to produce more convincing ones. In the original formulation this is a minimax game in which the discriminator tries to maximise, and the generator tries to minimise, the expected log-probability of classifying real and generated samples correctly. At the ideal equilibrium the generated data cannot be distinguished from the real data.

GANs are used for realistic image synthesis, super-resolution, image-to-image translation, style transfer, data augmentation for scarce medical or industrial data, and design exploration. Their strengths are sharp and realistic outputs and the ability to learn from unlabelled data. Their weaknesses are unstable training, a failure known as mode collapse in which the generator produces only a narrow range of outputs, the absence of a single reliable score for output quality, and ethical risks such as deepfakes. Diffusion models have recently overtaken GANs in many image-generation tasks, although GANs remain important for fast generation.

## *4.5 Comparative Analysis and Debate Verdict*

Table 4.1 places the three architectures against the same criteria, and Figure 4.2 turns the comparison into a practical guide for choosing between them. The debate ended without a single winner, because the architectures solve different problems. An RNN is best when the order of the data carries the meaning, a CNN when the meaning lies in spatial patterns, and a GAN when the aim is to create new data instead of predicting a label. In practice they are often combined, for example a CNN that extracts image features feeding an LSTM that writes a caption, or a conditional GAN guided by class labels.

The team also noted a difference in how the three are judged. RNNs and CNNs are normally evaluated with a clear metric on held-out data, such as accuracy, word error rate, or mean absolute error, whereas a GAN has no single loss that reflects the quality of its samples, so human inspection and scores such as the Fréchet Inception Distance are used. This makes GANs harder to engineer reliably even though their outputs can be the most striking of the three.

***Table 4.1:*** *Comparison of RNN, CNN, and GAN on the criteria used in the debate*

| Criterion | RNN | CNN | GAN |
|---|---|---|---|
| Typical input | Sequences: text, speech, time series | Grids: images, video frames, spectrograms | Noise vector plus real data |
| Core idea | Hidden state reused at each step | Shared filters and pooling | Generator versus discriminator |
| Main use | Prediction over sequences | Classification and detection | Generating new samples |
| Key strength | Memory of earlier inputs | Few parameters, strong features | Sharp, realistic outputs |
| Key weakness | Slow, sequential training | Needs large labelled data | Unstable training, mode collapse |
| Training difficulty | Vanishing gradients | Moderate | Highest |

***Table 4.2:*** *Landmark models in the history of each architecture*

| Architecture | Landmark model | Year | Contribution |
|---|---|---|---|
| RNN | LSTM (Hochreiter and Schmidhuber) | 1997 | Gates that preserve long-range memory. |
| RNN | GRU (Cho et al.) | 2014 | Simpler gated unit with fewer parameters. |
| CNN | LeNet-5 (LeCun et al.) | 1998 | First practical CNN for digit recognition. |
| CNN | AlexNet (Krizhevsky et al.) | 2012 | Won ImageNet and started the deep learning era in vision. |
| GAN | GAN (Goodfellow et al.) | 2014 | Adversarial training of a generator. |
| GAN | DCGAN (Radford et al.) | 2015 | Stable convolutional GAN design. |

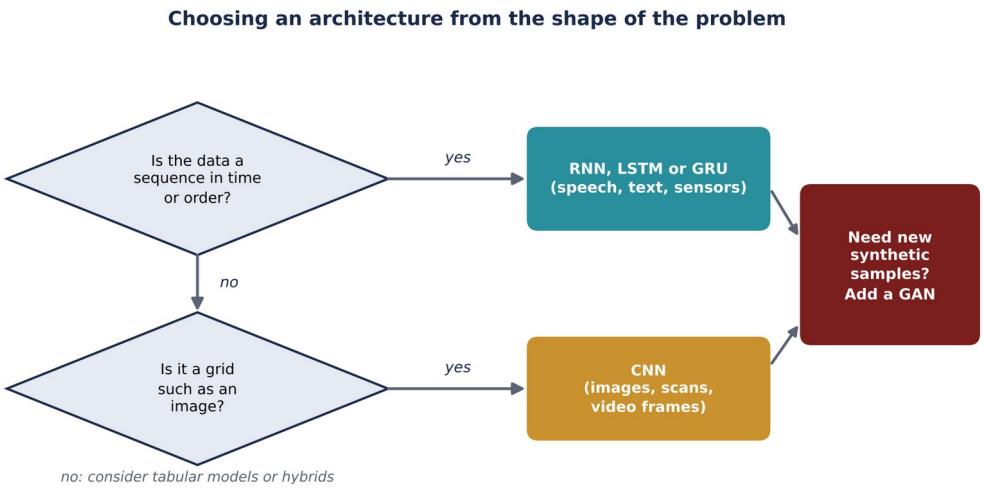



***Figure 4.2:*** *A simple guide for choosing between the three architectures.*

## *4.6 Conclusion*

The debate showed that each architecture is the right answer to a different question. Recurrent networks remember, convolutional networks see, and generative adversarial networks create. A good engineer therefore begins with the nature of the data and the goal of the task, not with a favourite model, and remains alert to newer designs such as Transformers and diffusion models that now handle many of the tasks once assigned to these three. Understanding the original architectures remains valuable because the ideas of memory, local feature extraction, and adversarial training continue to appear inside those newer models. The group also agreed that every architecture should be used responsibly: recurrent and convolutional models inherit the biases of their training data, as Module 3 showed, and generative models can be used to produce misleading content, so data governance, testing, and transparency apply to all three.

# MODULE 5
# MOCK VOICE-COMMERCE DESIGN

***Question (Module 5):*** *Mock Voice-Commerce Design: Teams design a voice-commerce flow (e.g., ordering groceries). They map user intents → voice recognition → intent parsing → response/output.*

## *5.1 Introduction and Objective*

Voice-commerce is the buying and selling of goods through spoken conversation with an assistant on a phone, a smart speaker, or a car system. It combines speech recognition, language understanding, dialogue management, and an online shop's back-end. Groceries are an attractive example because orders are frequent, repetitive, and made of short items. The objective was to design a grocery-ordering flow that maps user intent through voice recognition and intent parsing to a spoken response, and to examine its strengths and limits.

## *5.2 System Architecture and Workflow*

The design is a pipeline of four stages, shown in Figure 5.1. In the first stage the microphone captures speech, usually after a wake word, and a speech recognition model converts it to text. In the second stage a language-understanding component identifies the intent and extracts the entities, such as product and quantity. In the third stage the back-end looks up the product, checks stock, updates the cart, and, at checkout, verifies the user and authorises payment. In the fourth stage a reply is composed and converted to speech. A dialogue manager runs across all stages, remembering the conversation and asking questions when information is missing.

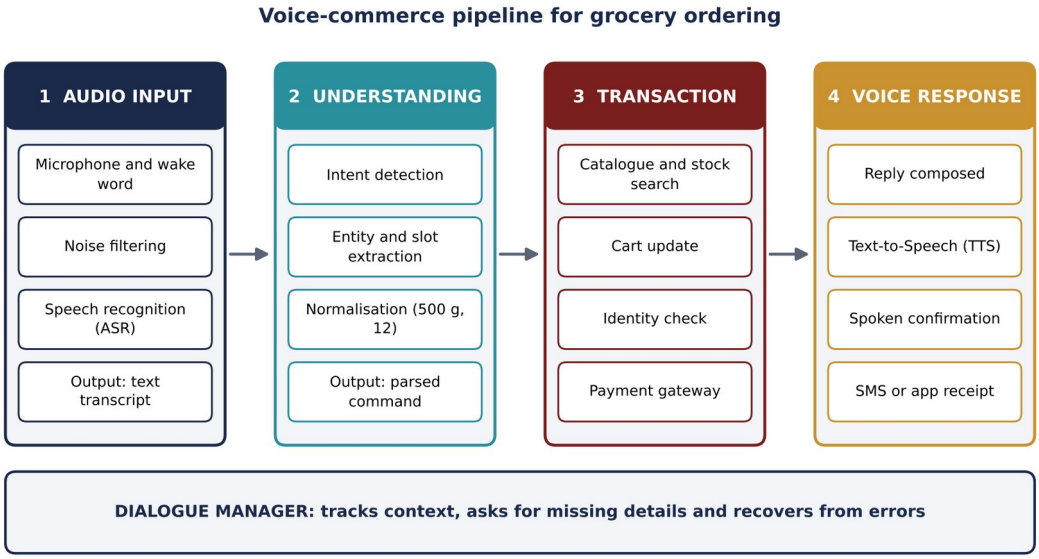



***Figure 5.1:*** *Voice-commerce pipeline for grocery ordering, from audio input to spoken response, with the dialogue manager running across the stages.*

## *5.3 Core Components*

Automatic speech recognition (ASR) converts audio into a transcript. An acoustic model based on a deep neural network maps spectral features to sounds, and a language model chooses the most probable word sequence. A vocabulary of grocery terms such as paneer, atta, and basmati improves accuracy on product names, and each transcript carries a confidence score so that doubtful results can be repeated back to the user instead of being acted upon.

Intent parsing, a task of natural language understanding, turns the transcript into a structured command. Intent classification decides what the user wants, for example to add an item, remove an item, change a quantity, view the cart, or check out, and entity extraction fills the slots that the action needs. Spoken forms are normalised so that “half a kilo” becomes 500 grams. Figure 5.2 shows the result for one command. The dialogue manager keeps track of the cart and the last item mentioned, so that a later phrase such as “make it three” can be resolved, asks for a missing quantity, and offers the best three matches when a word is ambiguous.

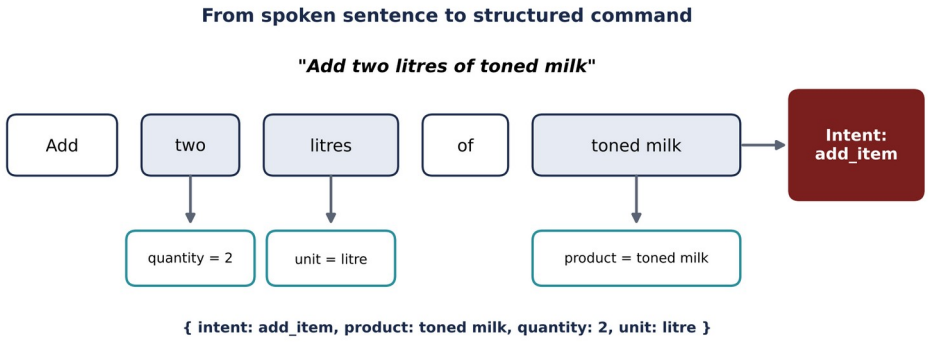



***Figure 5.2:*** *Intent parsing and entity extraction for a spoken grocery command.*

The last component is response generation. The assistant selects or composes a short reply, often from templates that are filled with the values in the cart, and a text-to-speech engine converts it to audio. Replies are kept brief, lists are limited to about three options, and every irreversible action is preceded by a read-back that the user must confirm.

## *5.4 Sample User Interaction*

Table 5.1 shows an abridged conversation in which the assistant, named Grocer for this mock design, handles a clarification, a correction, and a verified payment. All products and prices are sample data.

***Table 5.1:*** *Sample grocery-ordering conversation and the system action behind each turn*

| Speaker | Utterance | System action |
|---|---|---|
| User | “Hey Grocer, add two litres of toned milk.” | Wake word; intent add_item; product, quantity 2, unit litre. |
| Assistant | “Added two litres of toned milk. Anything else?” | Catalogue match found; cart updated. |
| User | “Get me some tomatoes.” | Intent add_item; quantity slot is missing. |
| Assistant | “How many kilos of tomatoes would you like?” | Dialogue manager asks for the slot. |
| User | “Half a kilo. And add bread.” | 500 g stored; bread has several matches. |
| Assistant | “I found white, brown, and multigrain bread. Which one?” | Top three matches offered. |
| User | “Brown, two packets. Actually, make the milk three litres.” | Slot filled; context resolves “the milk”; quantity updated. |

| Speaker | Utterance | System action |
|---|---|---|
| User | “Place the order.” | Intent checkout; read-back of items, total, and address. |
| Assistant | “Please say your voice PIN to confirm payment by UPI.” | Identity check before payment is released. |
| Assistant | “Payment successful. Your order arrives tomorrow morning.” | Order saved; SMS and app receipt sent. |

The conversation shows four design decisions: a missing slot is requested instead of guessed, an ambiguous product is resolved with a short list, a correction is applied through conversation memory, and payment is released only after read-back and verification. Each decision lowers the chance of a wrong order, which matters in voice because the user cannot scan the cart.

## *5.5 Challenges, Tools and Advantages*

The main challenges are listed here in the order the team judged them. Background noise and accent variation lower recognition accuracy, particularly with code-mixed speech and local product names, and are reduced by noise suppression, multilingual models, and confirmation of low-confidence words. Ambiguous commands are handled by clarifying questions and purchase history. Payment security is the most serious risk because a spoken command alone is weak proof of identity, so the design reads the order back, requires a voice PIN or one-time password, and applies spending limits. Privacy is protected by on-device wake-word detection, consent, and encryption.

The prototype can be built from common tools: Google Cloud Speech-to-Text, Amazon Transcribe, or the open-source Whisper model for recognition; spaCy or Rasa for language understanding and dialogue management; a Python framework such as FastAPI with PostgreSQL for the back-end; a payment gateway for UPI and cards; and a cloud text-to-speech service for replies. Quality is measured by word error rate, intent and slot accuracy, and task completion rate. The advantages are hands-free convenience, speed for repeat orders, accessibility for visually impaired and elderly users, and support for regional languages.

## *5.6 Overall Conclusion and Future Scope*

The four modules of this project follow one idea from the data upward. Module 2 showed how personal data can be misused and how governance can prevent harm. Module 3 showed how models trained on such data can learn bias and how fairness can be measured and improved. Module 4 compared the RNN, CNN, and GAN architectures for sequences, images, and generated data. Module 5 combined these ideas in a voice-commerce system, where privacy, fairness, and security decide whether users can trust the result.

Future work could apply the fairness audit to a real dataset, build the voice system, and test it on different accents and languages. Taken together, the work shows that practical artificial intelligence rests on three foundations at once: sound data practice, fair and well-understood models, and designs that respect the people who use them.

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